Video Transcript

Video 5: Reinforcement Learning: The Quality Matrix and the Bellman Equation (10:36)

So, if you remember last time, we were doing reinforcement machine learning and we were trying to capture the geometry of a problem and we managed to get the layout here in a graph form and we decided that we'd flag whether an action was possible or not by a one. If it was possible, we'd have a one. If it's not possible, then zero. And what we mean by an action is moving from one room to another. So, for example, moving from room one to five is possible. Moving from room one to two is not possible.

So, we started out with state zero, so down here, and the only possible move is to four. So, we put a one corresponding to that action of moving from zero to four. Now, you needed to complete being in state one. What were the possible actions? Well, you could go to three or five. So, this was the table that you needed to complete. This is what I asked you to do. So, this is the solution and this is what we call our reward matrix, R. Now, we're going to add a wrinkle to this that we're going to add for our goal 100 units as a reward.

So, we're going to update our reward so that if you move from one to five, you get a reward of 100. If you move from four to five, you get 100. And we also, we add this one in that if we stay at five move from five to five, it's 100. Now, it'll be clear why we need this one in as we move on. Now what we're going to do is we're going to introduce another matrix, we're going to call it a quality of move matrix or Q-matrix for short. And this is going to specify, these are going to be our bread crumbs.

So, these numbers are going to be high in some areas and low in other areas. So, we know what direction we need to move in. So, let's take a look at this move. The quality of moving from one to five. So, it means we're dealing with this row here one to five. Now this is the formula we're going to apply is that the quality of being in state S and moving to A depends on the reward of moving from S to A. So, moving from one to five, we have a reward here of 100. So, this is going to be at least a 100.

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Now, we choose gamma as being 0.5. I'm choosing 0.5 so that it makes the calculations easier. And now we look at where we move to the Q in the row of where we've moved to. So, we've moved to row five or state five. So, we look along this row and see if there are any Q values and we choose the largest. Now at the beginning, there aren't any here. So, we're just left with that, Q now will get updated to be 100, and this quantity is zero on the right. So, this becomes 100. Let's move on now.

So, the algorithm we're going to apply is we're going to choose a state at random, see what actions are possible, and calculate the Q value for one of those actions. So, let's we've seen Q for one to five. Let's now take a look at Q for one to three. So, here we are in one, and here's three. So, we know we've got a reward of one. Now it means that we've moved from one to three. Now, we need to look in row three to see if there's any other Q that contributes.

We're looking at this term here, this Q(s_next), s_next is three. And so we look at this row and there's no value there. So, we're just going to be left with that reward of one. So now, our Q gets updated in row one, and we've got a one here and 100 here. Let's go and see another action, three to one. So, here's three to one. We know we've got a reward of one there. Now let's see, we have to look in row one now to see if we've got any Q values. And we do. We've already got a one here and we've got 100, and we take the maximum one. So, we take the 100 multiplied by gamma, which is 0.5, so that's 50. So, and the one from our reward in our pocket, if you like, makes this 51. So, now we've got our Q value here of 51. Let's take another one, moving from two to three. So, here we are in two to three. So, we've got the reward of one. Now we need to look in row three to see what we've got there. Well, we've got a 51 there, so we take half of that because that's the maximum in that row. We take half of it, 25, I'm rounding down and then add that one.

So, we should get 26. You're beginning to see the pattern, feeling comfortable with this? Let's do another one, we'll do one to three again. Now this had a value of one already in there, but things have changed since then. So, now we need to take a look at the reward moving from one to three. So, that was a reward of one. Now, we look in row three and we see that the maximum Q in there is 51. So, we're going to get 25, when we multiply by the gamma, the half and adding one, this gets updated from one to 26. We'll just do a couple more, five to one. So, here we are. We're in row five, and we're going to move to row one. Now, five to one has a reward of one. So, this term, the R of the s, a is one. And now we need to look in row one to see what the

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maximum reward is, and it's 100. So, we're going to get 50 plus that one, 51. I'm just going to do one more. This is the last one, I promise you. Now, four to five. So, well, there's a reward of 100. So, this is already 100 of going from four to five. Now we need to look in row five to see the Q, maximum Q, and it's 51. So, we take half of that and add it to this. And this time I round it up. So, that gives 126. I just wanted to show you that we could, even though we start with 100 here, our maximum reward, this can increase because other numbers can contribute to it. So, we do this until this Q matrix is static and has converged and I ran it for 100 iterations and it converged to this. So, now our robot can find its way around.

Let's start the robot in row two. So, the robot starts in row two. It looks along here for actions that it can take, and it looks for the maximum quality and it sees a 52 here. So, it takes that. So, that means it's going to go to three. This action is move from two to state three. So, now we're in state three. We're here. Now, we're in this row and we look for the maximum action. In this case, we've got two that are the same. So, we could go to one or we could go to four and you could see that yes, those are good, those are equally possible and will get us to the goal. So, let's suppose we go to four, let's go to four first. So, we look along here and we see we've got two of 52s, but we've got the 200 here moving from four to five. That's the best quality move, and so we get to our goal. So, that's how Q-matrix, once we've got it, will help our robot find the goal. No matter where you put it, now it's got a recipe for getting to the goal. Let's see what it means in terms of code.

So, we're going to choose a random state. We're going to find a list of possible actions. Choose an action at random, scan that row of Qs and choose the maximum Q. Now, update the Q value using our formula, our Bellman's equation, so adding in whatever reward plus gamma times the maximum Q in that row. So, last time we ended up with our Q-matrix here, and we solved four to five. Now, I want you to update this Q-matrix as follows. I want you to find the solution of four to three and of zero to four. So, you need to find the Q values for both of these. Now make sure that you understand exactly what's going on because this is one of the most important things in reinforcement learning, understanding how we can create a map for the robot so it can find its way around. Okay, good luck. Bye for now.

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